

TRAIL VALLEY, NT, Canada

WMO: 716830

Lat: 68.742N

Lon: 133.499W

Elev: 85

StdP: 100.31

Time Zone: -7.00 (NAM)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-37.7	-35.9	-41.3	0.1	-37.4	-39.4	0.1	-35.7	18.3	-18.3	16.0	-18.3	1.9	0	0.910

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.7	24.6	15.6	22.3	14.9	20.0	14.1	16.7	22.6	15.6	20.7	14.5	19.1	4.6	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
14.4	10.4	18.6	13.3	9.6	17.4	12.2	8.9	16.4	47.4	22.7	44.1	20.9	40.9	19.0	21.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.3	10.3	8.8	DB	-40.3	28.3	2.0	2.0	-41.7	29.7	-42.8	30.8	-43.9	32.0	-45.3	33.4	(4)
(5)			WB	-40.1	18.9	2.2	1.1	-41.7	19.7	-42.9	20.3	-44.2	20.9	-45.8	21.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-7.7	-24.0	-23.3	-22.4	-12.8	-0.9	9.1	12.5	9.7	3.9	-5.7	-17.0	-22.4	(6)	
	DBStd	14.90	7.21	7.35	6.88	7.10	6.65	5.12	4.19	4.36	4.20	4.85	5.71	6.80	(7)	
	HDD10.0	6668	1055	933	1004	684	341	77	22	59	190	487	810	1004	(8)	
	HDD18.3	9507	1313	1166	1262	934	595	278	185	270	434	745	1060	1263	(9)	
	CDD10.0	212	0	0	0	0	4	51	100	50	6	1	0	0	(10)	
	CDD18.3	9	0	0	0	0	0	2	5	2	0	0	0	0	(11)	
	CDH23.3	121	0	0	0	0	4	43	55	19	0	0	0	0	(12)	
	CDH26.7	18	0	0	0	0	1	8	7	2	0	0	0	0	(13)	
(14) Wind	WSAvg	3.8	3.9	4.1	3.8	3.6	3.9	4.2	4.1	3.9	3.8	3.4	3.3	3.2	(14)	
(15) Precipitation	PrecAvg	239	11	10	13	10	12	25	40	40	31	23	14	12	(15)	
	PrecMax	314	30	20	34	26	36	69	73	81	56	42	34	35	(16)	
	PrecMin	181	1	0	0	1	0	3	10	13	8	5	1	2	(17)	
	PrecStd	36	8	6	8	7	8	17	15	16	12	9	7	8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-0.3	-2.8	-0.2	6.4	20.7	27.9	27.6	26.0	18.7	8.7	-0.3	-3.5	(19)	
		MCWB	-1.1	-3.6	-1.2	3.6	11.4	16.8	17.0	17.7	12.0	6.0	-0.6	-3.9	(20)	
	2%	DB	-5.1	-5.9	-5.5	3.3	16.3	24.0	24.8	22.1	15.2	3.9	-4.1	-6.6	(21)	
		MCWB	-5.6	-6.6	-6.3	1.3	9.1	14.9	15.7	15.8	10.2	2.7	-4.5	-7.0	(22)	
	5%	DB	-9.6	-8.7	-9.4	0.7	12.2	21.5	22.7	19.4	12.7	2.0	-6.4	-10.1	(23)	
		MCWB	-10.0	-9.2	-10.1	-0.7	7.3	13.5	15.4	14.6	9.2	1.1	-6.8	-10.3	(24)	
	10%	DB	-14.1	-12.1	-12.5	-1.9	8.6	18.7	20.6	17.1	10.1	0.4	-8.7	-12.8	(25)	
		MCWB	-14.4	-12.4	-13.1	-3.0	5.3	12.3	14.8	13.3	8.0	-0.1	-9.0	-13.1	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-2.1	-3.7	-1.1	2.4	11.1	17.4	18.1	18.0	12.7	7.3	-0.5	-3.6	(27)	
		MCDB	-1.5	-3.1	0.2	4.1	18.4	25.6	23.8	24.5	16.5	9.5	-0.2	-3.2	(28)	
	2%	WB	-5.6	-6.5	-5.8	0.3	8.7	15.5	17.0	16.4	11.0	3.1	-4.0	-6.9	(29)	
		MCDB	-5.1	-5.9	-5.1	1.8	14.9	22.7	22.8	21.2	14.3	4.0	-3.7	-6.5	(30)	
	5%	WB	-9.5	-9.0	-9.5	-1.4	6.8	14.1	16.0	15.0	9.5	1.4	-6.3	-10.3	(31)	
		MCDB	-9.2	-8.4	-8.8	-0.2	11.2	20.7	21.1	18.3	12.3	2.1	-6.0	-10.0	(32)	
	10%	WB	-13.7	-12.2	-12.8	-3.7	4.9	12.6	15.0	13.5	8.0	0.0	-8.6	-12.9	(33)	
		MCDB	-13.4	-11.8	-12.2	-2.7	7.8	18.1	19.7	16.5	10.3	0.6	-8.2	-12.7	(34)	
(35) Mean Daily Temperature Range	MDBR		6.3	7.0	7.4	8.6	8.5	10.8	9.7	8.3	6.5	5.1	5.9	6.1	(35)	
	5% DB	MCDBR	10.2	8.8	8.9	9.2	14.4	15.0	13.6	12.1	10.7	6.2	6.6	9.2	(36)	
		MCWBR	9.5	8.3	8.1	7.9	8.6	7.7	6.7	6.6	6.2	5.0	6.5	9.0	(37)	
	5% WB	MCDBR	9.9	8.6	8.7	9.0	13.3	14.1	12.1	11.0	10.2	6.1	6.6	9.4	(38)	
		MCWBR	9.5	8.3	8.0	7.6	8.2	7.7	6.6	6.5	6.3	4.9	6.5	9.3	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.157	0.212	0.250	0.302	0.336	0.338	0.360	0.334	0.298	0.233	0.157	N/A	(40)	
	taud		1.908	2.111	2.221	2.173	2.315	2.457	2.439	2.520	2.545	2.217	1.917	N/A	(41)	
	Ebn at Noon		90	595	795	838	850	862	825	812	755	551	72	0	(42)	
	Edh at Noon		6	42	72	104	103	93	92	76	56	37	5	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.03	0.49	2.12	4.42	5.85	6.21	5.01	3.35	1.79	0.58	0.08	0.00	(44)	
	RadStd		0.00	0.04	0.10	0.18	0.41	0.48	0.47	0.34	0.16	0.07	0.01	0.00	(45)	

Historical Trends

		Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (7 neighbors)		+1.04	+1.81	+2.05	N/A	N/A	N/A	-427	-422	N/A	N/A

Nomenclature: See separate page